

The Ethical Hacker's Command Handbook

Chapter 2 – Part 3: Port Selection & Service Version Detection

Knowing which ports to scan and identifying the services running on them are key steps in authorized network enumeration.

Learning Objectives

Learn how to scan specific ports, multiple ports, ranges, all TCP ports, perform fast scans, use top-port scans, detect service versions, and understand when each option is most appropriate.

Why Port Selection Matters

By default, Nmap scans the 1,000 most common TCP ports. While this is efficient, important services may run on uncommon ports, making targeted or full-port scans necessary in some assessments.

nmap -p 80

Purpose: Scan one specific port

What it does: Checks only the selected port.

When to use: When verifying a known service or troubleshooting.

Avoid when: Initial reconnaissance when you don't know which ports are open.

Memory Tip: Remember: -p = Port.

nmap -p 22,80,443

Purpose: Scan multiple ports

What it does: Scans selected ports only.

When to use: Web server and SSH assessments.

Avoid when: When full enumeration is required.

Memory Tip: Separate ports with commas.

nmap -p 1-1000

Purpose: Scan a port range

What it does: Scans a continuous range of ports.

When to use: Focused assessments of lower-numbered ports.

Avoid when: When uncommon ports may be in use.

Memory Tip: Use ranges to save time.

nmap -p-

Purpose: Scan all TCP ports

What it does: Scans all 65,535 TCP ports.

When to use: CTFs, TryHackMe, thorough assessments.

Avoid when: Quick reconnaissance.

Memory Tip: Hyphen means all ports.

nmap -F

Purpose: Fast scan

What it does: Scans fewer ports than the default.

When to use: Quick overview of a host.

Avoid when: When comprehensive results are required.

Memory Tip: F = Fast.

nmap --top-ports 100

Purpose: Top ports scan

What it does: Scans the most common ports.

When to use: Large environments or time-limited assessments.

Avoid when: When unusual ports are suspected.

Memory Tip: Choose the number based on your goal.

nmap -sV

Purpose: Version detection

What it does: Identifies software and service versions.

When to use: Immediately after discovering open ports.

Avoid when: When only checking host availability.

Memory Tip: V = Version.

nmap -sV --version-light

Purpose: Light version detection

What it does: Uses fewer probes for faster results.

When to use: When speed is more important than detail.

Avoid when: When exact versions are critical.

Memory Tip: Faster but less thorough.

nmap -sV --version-all

Purpose: Full version detection

What it does: Uses additional probes for better identification.

When to use: When accurate version information is essential.

Avoid when: Quick reconnaissance.

Memory Tip: More accurate but slower.

Typical Enumeration Workflow

1. Scan ports (-sS)
2. If necessary, scan all ports (-p-)
3. Detect service versions (-sV)
4. Research discovered services
5. Continue with authorized enumeration using appropriate NSE scripts.

Command Summary

Goal	Command
One port	-p 80
Multiple ports	-p 22,80,443
Port range	-p 1-1000
All ports	-p-
Fast scan	-F
Top ports	--top-ports 100
Version detection	-sV
Fast version detection	--version-light
Detailed version detection	--version-all

Revision Notes

- -p → Specific ports
- -p- → All TCP ports
- -F → Fast scan
- --top-ports → Most common ports
- -sV → Version detection
- --version-light → Faster version detection
- --version-all → More detailed version detection

Practice Questions

1. What is the difference between -p 80 and -p-?
2. Why might a default scan miss an important service?
3. When would you use -F?
4. What information does -sV provide?
5. When is --version-all preferable to the default -sV?